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An improved condition for the convexity of Bernstein–Bézier surfaces over triangles

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Abstract. A sufficient condition which is superior to that of Chang and Davis for the convexity of the Bernstein–Bézier polynomials of degree n over triangles is presented. The condition is proved to be necessary also for $n = 2$ and $n = 3$.

Keywords. Bernstein–Bézier polynomials over triangles, Bézier ordinates, convexity.

1. Introduction

Bernstein–Bézier triangular patches have recently and extensively studied in the context of CAGD [Sabin 1977; Farin 1979, 1983; Barnhill and Farin 1981; Goldman 1983].

Let T be a given triangle with vertices T_1, T_2, T_3. For point $p \in T$, let (u, v, w) be barycentric coordinates of p with respect to T. It is well-known that $0 \leqslant u, v, w \leqslant 1$ and $u + v + w = 1$. To each set $f = \{f_{i,j,k}: i, j, k$ are nonnegative integers such that $i + j + k = n\}$ the nth Bernstein–Bézier polynomial

$$B_n(f; p) = \sum_{i+j+k=n} f_{i,j,k} J^n_{i,j,k}(p) \tag{1}$$

is associated, in which

$$J^n_{i,j,k}(p) = \frac{n!}{i!j!k!} u^i v^j w^k, \quad i + j + k = n. \tag{2}$$

$f = \{f_{i,j,k}\}$ is called the set of Bézier ordinates of the polynomial (1). Geometrically (1) represents a triangular patch over T. For more detail we refer the readers to [Farin 1979, 1983].

Chang and Davis [Chang and Davis 1984] initiate the investigations of convexity of the Bernstein–Bézier triangular patches. The function $F(p)$ is said to be convex over T if the following inequality

$$F\left(\frac{p_1 + p_2}{2}\right) \leqslant \tfrac{1}{2}[F(p_1) + F(p_2)]$$

holds for any p_1, p_2 in T. According to [Chang and Davis 1984], the set f is said to be convex

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in 3-direction, if the Bézier ordinates satisfy

$$\begin{aligned} \Delta^{(1)}_{i,j,k} &= (E_2 - E_1)(E_3 - E_1) f_{i,j,k} \geqslant 0, \\ \Delta^{(2)}_{i,j,k} &= (E_3 - E_2)(E_1 - E_2) f_{i,j,k} \geqslant 0, \\ \Delta^{(3)}_{i,j,k} &= (E_1 - E_3)(E_2 - E_3) f_{i,j,k} \geqslant 0 \end{aligned} \tag{3}$$

for all i, j, k, such that $i+j+k=n-2$, where E_1, E_2 and E_3 are characterized by

$$\left.\begin{aligned} E_1 f_{i,j,k} &= f_{i+1,j,k} \\ E_2 f_{i,j,k} &= f_{i,j+1,k} \\ E_3 f_{i,j,k} &= f_{i,j,k+1} \end{aligned}\right\} \quad \text{for } i+j+k=n-1.$$

The above algebraic definition of 'convex in 3-direction' can be equivalently described geometrically as 'each pair of adjacent subtriangles in the Bézier net forms a convex surface'. For the definition of Bézier net, the reader may consult the papers by Farin, just mentioned above.

The authors of present paper showed in an unpublished paper (1983) that f is convex in 3-direction if and only if the corresponding Bézier net is convex over T.

One of the main theorems due to [Chang and Davis 1984] is that if f is convex in 3-direction, then $B_n(f;\ p)$ is convex over T.

Condition (3) is sufficient rather than necessary for the convexity of $B_n(f;p)$ over T. Let us consider the following example: a Bernstein–Bézier triangular patch is defined by the following data: all the Bézier ordinates are zero except

$$f_{4,0,0} = f_{0,4,0} = f_{0,0,4} = 1 \quad \text{and} \quad f_{0,2,2} = f_{2,0,2} = f_{2,2,0} = \tfrac{1}{3}.$$

The corresponding Bernstein–Bézier polynomial is

$$B_4(f;p) = (u^2 + v^2 + w^2)^2. \tag{4}$$

Even though in this case f is not convex in 3-direction, one can verify without difficulty that $B_4(f;p)$ is convex over T.

In this paper another condition which is weaker than (3) for the convexity of $B_n(f;p)$ is provided. It will be shown that the condition is not only sufficient but also necessary for the convexity of $B_n(f;p)$ in the cases of $n=2$ and $n=3$.

2. Main results

Combination (12) and theorem 3 both in [Chang and Davis 1984] gives that $B_n(f;p)$ is convex over T iff the quadratic form

$$[\xi, \eta, \zeta] \begin{bmatrix} \dfrac{\partial^2 \mathscr{B}}{\partial u^2} & \dfrac{\partial^2 \mathscr{B}}{\partial u \partial v} & \dfrac{\partial^2 \mathscr{B}}{\partial u \partial w} \\ \dfrac{\partial^2 \mathscr{B}}{\partial u \partial v} & \dfrac{\partial^2 \mathscr{B}}{\partial v^2} & \dfrac{\partial^2 \mathscr{B}}{\partial v \partial w} \\ \dfrac{\partial^2 \mathscr{B}}{\partial u \partial w} & \dfrac{\partial^2 \mathscr{B}}{\partial v \partial w} & \dfrac{\partial^2 \mathscr{B}}{\partial w^2} \end{bmatrix} \begin{bmatrix} \xi \\ \eta \\ \zeta \end{bmatrix} \geqslant 0 \tag{5}$$

for all $p \in T$ and all (ξ, η, ζ) such that $\xi + \eta + \zeta = 0$. In (5), $B_n(f;p)$ has been replaced by $\mathscr{B}$ for the sake of simplicity. Note that the matrix in (5) is the Hessian of $\mathscr{B}$. It should be pointed

out that (ξ, η, ζ) such that $\xi+\eta+\zeta=0$ represents a direction in barycentric coordinates and that the left-hand side is just the second directional derivative of $\mathscr{B}$ with respect to the direction (see [Farin 1979, 1983] again). Simple calculations yield

$$\frac{\partial^2\mathscr{B}}{\partial u^2}=n(n-1)\sum_{i+j+k=n-2} E_1^2 f_{i,j,k} J_{i,j,k}^{n-2}(p),$$

$$\frac{\partial^2\mathscr{B}}{\partial v\partial w}=n(n-1)\sum_{i+j+k=n-2} E_2E_3 f_{i,j,k} J_{i,j,k}^{n-2}(p)$$

etc. (5) can be rewritten as

$$\sum_{i+j+k=n-2} Q_{i,j,k}(\xi,\eta,\zeta)J_{i,j,k}^{n-2}(p)\geqslant 0 \tag{6}$$

where the quadratic form Q_{ijk} is defined by

$$Q_{i,j,k}=[\xi,\eta,\zeta]\begin{bmatrix} E_1^2 f_{i,j,k} & E_1E_2 f_{i,j,k} & E_1E_3 f_{i,j,k}\\ E_1E_2 f_{i,j,k} & E_2^2 f_{i,j,k} & E_2E_3 f_{i,j,k}\\ E_1E_3 f_{i,j,k} & E_2E_3 f_{i,j,k} & E_3^2 f_{i,j,k}\end{bmatrix}\begin{bmatrix}\xi\\ \eta\\ \zeta\end{bmatrix}, \tag{7}$$
$$\text{for } i+j+k=n-2.$$

Let us begin with $n=2$. In this case (6) becomes

$$[\xi,\eta,\zeta]\begin{bmatrix} A & c & b\\ c & B & a\\ b & a & C\end{bmatrix}\begin{bmatrix}\xi\\ \eta\\ \zeta\end{bmatrix}\geqslant 0 \tag{8}$$

where

$$\begin{aligned} &A=f_{2,0,0}, \quad B=f_{0,2,0}, \quad C=f_{0,0,2},\\ &a=f_{0,1,1}, \quad b=f_{1,0,1}, \quad c=f_{1,1,0}. \end{aligned} \tag{9}$$

It has been shown that (8) holds for $\xi+\eta+\zeta=0$ iff the 2×2 matrix

$$\begin{bmatrix} A+C-2b & C+c-a-b\\ C+c-a-b & B+C-2a\end{bmatrix} \tag{10}$$

is positive semidefinite. With

$$\begin{aligned} \Delta_a&=A+a-b-c,\\ \Delta_b&=B+b-c-a,\\ \Delta_c&=C+c-a-b, \end{aligned} \tag{11}$$

then the matrix in (10) reduces to

$$\begin{bmatrix} \Delta_a+\Delta_c & \Delta_c\\ \Delta_c & \Delta_b+\Delta_c\end{bmatrix}. \tag{12}$$

The matrix in (12) is positive semidefinite iff

$$\Delta_a+\Delta_c\geqslant 0, \qquad \Delta_b+\Delta_c\geqslant 0,$$
$$(\Delta_a+\Delta_c)(\Delta_b+\Delta_c)-\Delta_c^2\geqslant 0,$$

or, more symmetrically but equivalently,

$$\begin{aligned} &\Delta_b+\Delta_c\geqslant 0, \quad \Delta_c+\Delta_a\geqslant 0, \quad \Delta_a+\Delta_b\geqslant 0,\\ &\Delta_b\Delta_c+\Delta_c\Delta_a+\Delta_a\Delta_b\geqslant 0. \end{aligned} \tag{13}$$

Hence we have proven the following:

Theorem 1. *$B_2(f;p)$ is convex over T iff* (13) *holds, where Δ_a, Δ_b, Δ_c are given by* (11) *with quantities defined in* (9).

Theorem 2. *$B_3(f;p)$ is convex over* T *iff*

$$\begin{aligned}&\Delta^{(2)}_{i,j,k}+\Delta^{(3)}_{i,j,k}\geqslant 0,\quad \Delta^{(3)}_{i,j,k}+\Delta^{(1)}_{i,j,k}\geqslant 0,\quad \Delta^{(1)}_{i,j,k}+\Delta^{(2)}_{i,j,k}\geqslant 0,\\&\Delta^{(2)}_{i,j,k}\Delta^{(3)}_{i,j,k}+\Delta^{(3)}_{i,j,k}\Delta^{(1)}_{i,j,k}+\Delta^{(1)}_{i,j,k}\Delta^{(2)}_{i,j,k}\geqslant 0\end{aligned}\tag{14}$$

for $i+j+k=1$.

Proof. If $n=3$, then (6) becomes

$$uQ_{1,0,0}(\xi,\eta,\zeta)+vQ_{0,1,0}(\xi,\eta,\zeta)+wQ_{0,0,1}(\xi,\eta,\zeta)\geqslant 0.\tag{15}$$

As has been shown before, condition (14) implies that

$$\begin{aligned}&Q_{1,0,0}(\xi,\eta,\zeta)\geqslant 0,\\&Q_{0,1,0}(\xi,\eta,\zeta)\geqslant 0,\\&Q_{0,0,1}(\xi,\eta,\zeta)\geqslant 0,\end{aligned}\tag{16}$$

for all (ξ,η,ζ) such that $\xi+\eta+\zeta=0$. Since the left-hand side of (15) is convex linear combination of $Q_{1,0,0}$, $Q_{0,1,0}$, $Q_{0,0,1}$, (15) holds for each $(u,v,w)\in T$ and $\xi+\eta+\zeta=0$. Hence $B_3(f;p)$ is convex over T.

Conversely, we assume that $B_3(f;p)$ is convex over T, thus (15) holds for each $(u,v,w)\in T$. Taking in turn $u=1$, $v=1$ and $w=1$, we obtain (16) from (15). Hence the condition (14) must hold. □

Generally, for each $p\in T$, the left-hand side of (6) is convex linear combination of $Q_{i,j,k}(\xi,\eta,\zeta)$ in which $i+j+k=n-2$ and $\xi+\eta+\zeta=0$. Now follows immediately

Theorem 3. *If* (14) *holds for all $i+j+k=n-2$, then $B_n(f;p)$ is convex over T.*

Since it is obvious that (3) implies (14), we get as a special case of Theorem 3 the following

Corollary [Chang and Davis, 1984]. *If f is convex in 3-direction, then $B_n(f;p)$ is convex over T.*

Given

$$\begin{aligned}&f_{2,0,0}=f_{0,2,0}=-f_{1,1,0}=1,\\&f_{0,0,2}=f_{0,1,1}=f_{1,0,1}=0,\end{aligned}$$

the associated $B_2(f;p)$ is convex over T as it is justified by Theorem 1. But Chang and Davis' criterion fails to be valid as the set of Bézier ordinates is not convex in 3-direction. Hence we see that the present criterion is superior to that of Chang and Davis.

Note that for $n\geqslant 4$ the condition (14) is still a sufficient rather than a necessary condition for the convexity of $B_n(f;p)$. The example in the introductory part of this paper also works to provide a counter-example.

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